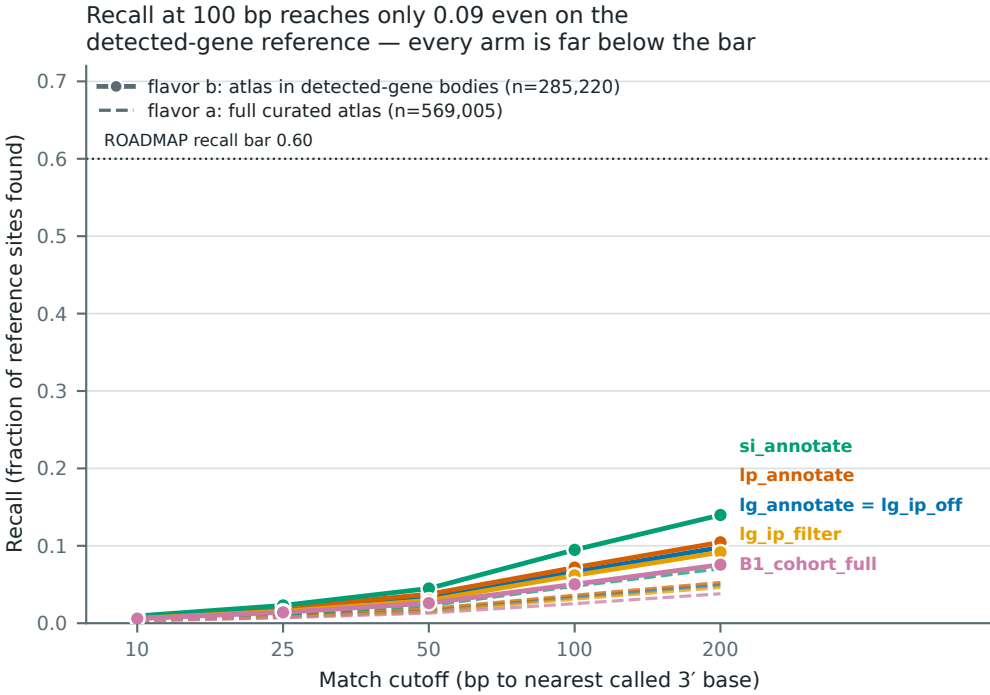
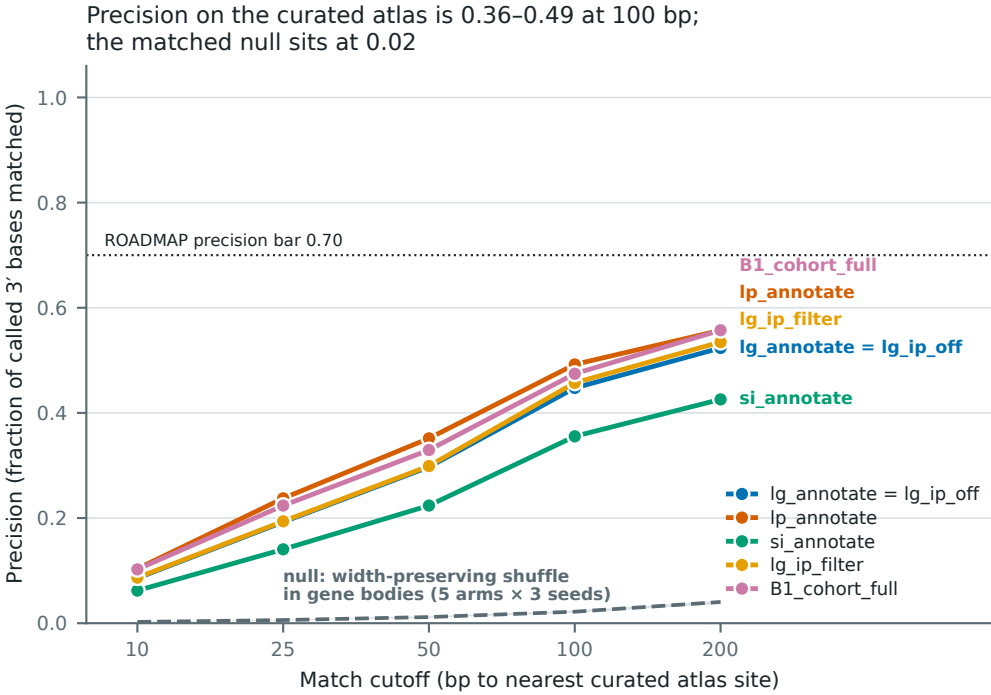


No arm clears any ROADMAP bar on the curated PolyASite 2.0 atlas: precision 0.36–0.49 at 100 bp (≈20× null), recall ≤ 0.09



Scorecard at 50 and 100 bp against the ROADMAP publication bars ($P \geq 0.70$, $R \geq 0.60$, $F1 \geq 0.65$ at the 100 bp malleable cutoff)

	at 50 bp				at 100 bp (roadmap cutoff)				TES P	bars @100 (flavor b)
	P	R (a)	R (b)	F1 (b)	P	R (a)	R (b)	F1 (b)		
lg_annotate = lg_ip_off	0.297	0.016	0.032	0.058	0.447	0.033	0.066	0.115	0.280	P X R X F1 X
n = 22,629 called 3' bases										
lp_annotate	0.352	0.019	0.037	0.068	0.492	0.036	0.072	0.125	0.313	P X R X F1 X
n = 21,470 called 3' bases										
si_annotate	0.224	0.023	0.045	0.075	0.355	0.048	0.095	0.150	0.204	P X R X F1 X
n = 43,023 called 3' bases										
lg_ip_filter	0.299	0.015	0.029	0.054	0.457	0.031	0.062	0.109	0.293	P X R X F1 X
n = 20,605 called 3' bases										
B1_cohort_full	0.329	0.013	0.026	0.048	0.475	0.025	0.050	0.091	0.303	P X R X F1 X
n = 16,497 called 3' bases										

Verdict at the 100 bp roadmap cutoff: 0/5 arms clear the precision bar; 0/5 clear recall and 0/5 clear F1 even on the favourable detected-gene reference (flavor b). Recall flavor (a), the full curated atlas, is lower still — an arm calling ~25k sites cannot recover a 569k-site atlas.

Matching: strand-matched ``bedtools closest -s -d`` between each called peak's 3'-most base ('+' end-1; '-' start) and the PolyASite 2.0 representative-site point BED (569,005 curated clusters, GRCh38.96) — point mode only; this is stricter than the pipeline's own strand-agnostic whole-interval window matcher, and none of these numbers is comparable to the 0.9986 previously quoted against the 18.4M-entry dump (see `null_control`). Null: 3 seeds of width-preserving ``bedtools shuffle -chrom -noOverlapping`` inside merged annotated gene bodies (1.80 Gb), reduced to the 3' base and scored identically; band = min-max over all arms × seeds. Recall flavor (a) denominator is all 569,005 curated sites and includes 397 sites on scaffolds no call can reach; flavor (b) restricts to sites inside strand-matched gene bodies of the 14,851 genes detected in this cohort (n=285,220). F1 pairs precision on the full curated atlas with the stated recall flavor. TES P = precision vs the 73,439 protein-coding transcript ends (Ensembl 99), a coarser sanity reference. `lg_ip_off/pasbed.bed` is byte-identical to `lg_annotate` (md5 20cb4550...), scored once. Scaffold peaks dropped per arm: `lg_annotate` 4, `lp_annotate` 4, `si_annotate` 12, `lg_ip_filter` 4, `B1_cohort_full` 3.